# MusicID: how it works, with all the maths

A song-recognition system built from first principles: a landmark fingerprint (after Wang, 2003), a statistically derived match decision, a neural fingerprint trained with contrastive learning, and a calibrated learned fusion, all benchmarked on 4,000 Creative-Commons songs (FMA). Full details and every experiment: MusicID_Technical_Report.pdf.

| **1.** Record<br>mic, 44.1 kHz | **2.** Spectrogram<br>resample + STFT | **3.** Peaks<br>constellation map | **4.** Hashes<br>target zones | **5.** Votes<br>time offsets | **6.** Decision<br>p-value / conformal |
|---|---|---|---|---|---|

## 1. From sound to a spectrogram

Audio at any sample rate is averaged to mono and resampled to $F_s$ = 11,025 Hz with a polyphase anti-aliasing filter (Nyquist 5,512.5 Hz). It is cut into frames of N = 1,024 samples every H = 512 samples (46.4 ms), each multiplied by a Hann window to reduce spectral leakage, and Fourier-transformed (FFT, O(N log N)). The log magnitude is kept:

$$X_t[k] = \sum_{n=0}^{N-1} x[tH+n]\, w[n]\, e^{-2\pi i k n/N}, \quad w[n] = \tfrac{1}{2}\left(1-\cos\tfrac{2\pi n}{N}\right), \quad S_t[k] = \log|X_t[k]|, \quad \Delta f = \tfrac{F_s}{N} = 10.77 \text{ Hz}$$

## 2. Peaks and hashes

A point (t, k) is a peak if it is the largest value in a 7-frame × 21-bin neighbourhood and above the median; at most 30 peaks per second are kept (215 Hz–5.4 kHz). These loud points survive noise and volume changes; together they form the **constellation map**. Each peak (anchor a) is paired with up to 6 later peaks p in its **target zone** ($1 \le t_p - t_a \le 40$ frames, $|k_p - k_a| \le 100$ bins), giving a 24-bit hash stored with the song id and anchor time:

$$h = k_a \cdot 2^{15} + k_p \cdot 2^6 + (t_p - t_a), \qquad \text{index}:\ h \mapsto \{(\text{song}, t_a)\}\ \ (\text{sorted arrays, binary search})$$

## 3. Votes: the time offset gives the match away

Every query hash found in the index votes for (song, δ) with $\delta = t_a^{\text{db}} - t_a^{\text{query}}$. For the right song all votes share one δ (where the clip starts in the song); random matches scatter. Each anchor votes once per (song, δ), and a song's score is its best window of three neighbouring offsets:

$$c_s = \max_{\delta}\left(N_{s,\delta-1} + N_{s,\delta} + N_{s,\delta+1}\right)$$

## 4. Deciding: a threshold with a known error rate

**Null model.** If the clip is not from song s, its $M_s$ votes fall at random over $K_s$ possible offsets, so a window count is about Poisson($w\lambda_s$), $\lambda_s = M_s/K_s$, w = 3. The top score must beat every offset of every song, so its family-wise p-value is

$$p(c) \approx 1 - \exp\left(-\sum_s K_s \Pr\left[\text{Poisson}(w\lambda_s) \ge c\right]\right), \qquad \text{match if } p(c) < \alpha = 1\%$$

**Over-dispersion.** Real votes come in clumps (a repeated musical pattern aligns many hashes at once), so the plain model promised 1% but gave 13.0% false matches. Quasi-Poisson divides counts by ϕ, estimated on held-out non-library clips (ϕ = 1.5): $c/\phi \sim \text{Poisson}(w\lambda_s/\phi)$. **Conformal:** for any score, the threshold $\tau = s_{(k)}$, k = (1−α)(n+1), of n calibration non-library clips guarantees P(false match) ≤ α (one threshold per clip length). **Sequential:** the app checks every second; using α/T at each of T checks keeps the total ≤ α (Bonferroni): $P(\text{t: } p_t < \alpha/T) \le \Sigma\, \alpha/T = \alpha$.

## 5. Neural fingerprint (contrastive learning)

A small CNN maps each 1-s log-mel segment to a unit vector f(x) of 128 numbers. It is trained on 2,992 songs not in the test library: pairs $(x, x^+)$ are a clean segment and a degraded copy (shift, noise, echo, EQ, ±3% speed), and the NT-Xent loss makes them the most similar pair in the batch:

$$\ell_i = -\log \frac{e^{f(x_i)^\top f(x_i^+)/\tau}}{\sum_{j \ne i} e^{f(x_i)^\top f(x_j)/\tau}}, \qquad \tau = 0.05$$

Search: nearest neighbours of each query segment among 228,000 library segments, then offset voting as in section 3.

## 6. Learned fusion

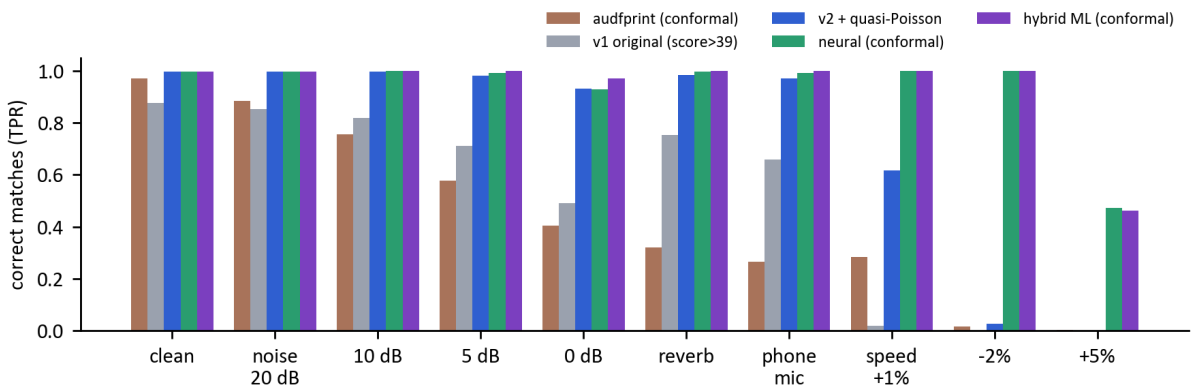
Two logistic regressions estimate the probability that the classical or the neural candidate is right from 13 evidence features x (−log p, scores, margins, agreement, clip length); the more probable candidate is reported if it passes the conformal threshold:

$$P(\text{right} \mid x) = \sigma(\beta^\top x) = \frac{1}{1+e^{-\beta^\top x}}, \qquad \hat{\beta} = \arg\min_{\beta} -\sum_i \omega_i \left[y_i \log \sigma_i + (1-y_i)\log(1-\sigma_i)\right]$$

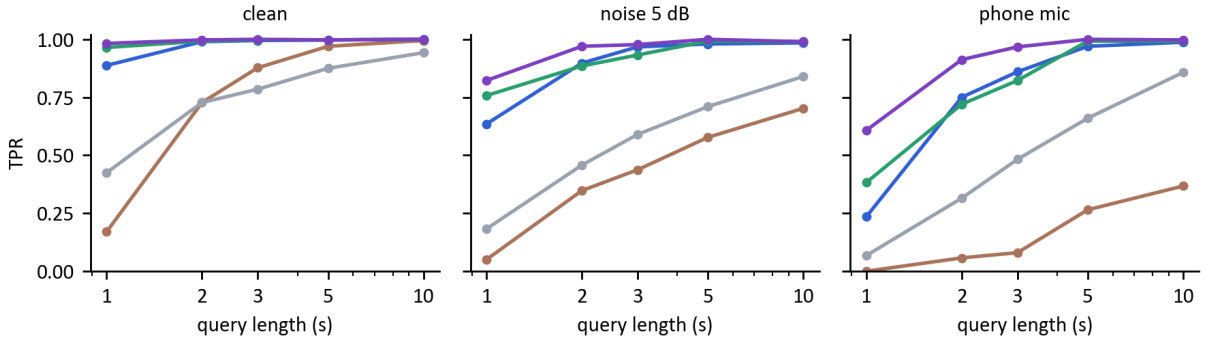
## 7. Results (4,000-song library, 17,600 test clips, 1% false-match target)

| Method | Correct matches | False matches (95% CI) | Top-1 |
|---|---|---|---|
| Original MusicID (score > 39) | 53.9% | 21.4% (19.2%–23.9%) | 53.9% |
| audfprint, conformal 1% | 42.2% | 0.5% (0.3%–0.7%) | 59.4% |
| v2 classical, conformal 1% | 81.2% | 1.2% (0.9%–1.7%) | 85.8% |
| Neural fingerprint, conformal 1% | 90.1% | 1.6% (1.0%–2.1%) | 98.0% |
| Hybrid (classical + neural, learned), conformal 1% | 93.9% | 1.4% (1.0%–1.8%) | 98.1% |

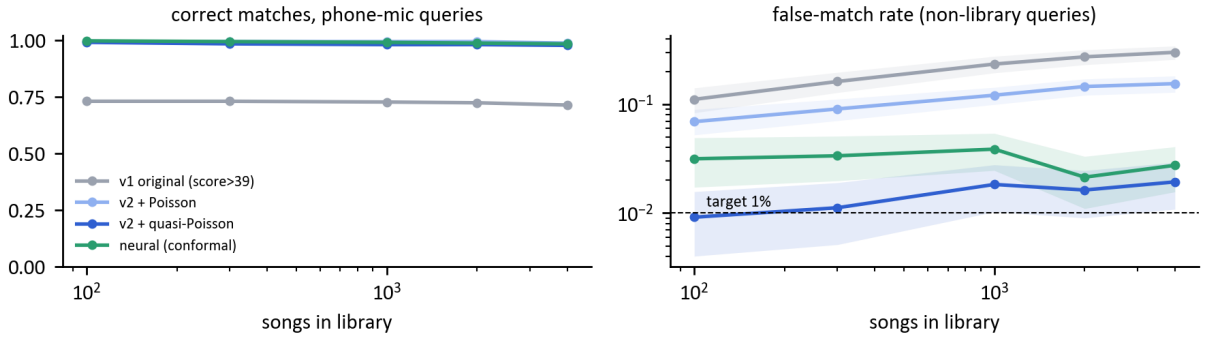
*Ten recording conditions (clean, noise 20–0 dB, reverb, phone microphone, speed ±1–5%) × clip lengths 1–10 s; the open-source audfprint is the external baseline; CIs bootstrap over songs.*




*Correct matches for 5-s clips by recording condition.*




*Correct matches vs clip length: the hybrid needs about 2 s even through a simulated phone microphone.*




*Library size: the guessed rule makes more false matches with every song added; the derived threshold stays near the 1% target (shaded: 95% CI).*

## 8. What the original version taught

An exact (bit-identical, unit-tested) replica of the first version showed that its guessed rule "score > 39" names a song for 21.4% of clips that are not in the library, rising with library size, and that its hand-written FFT is correct to $10^{-9}$. Its 40,001-tap anti-aliasing filter could be 157 taps: content above 11,025 − 5,000 = 6,025 Hz is all that can alias onto the bins used, and the shorter filter gives 98.5% identical peaks.

*Limits: simulated degradations, 30-s songs, a 4,000-song library and a small CPU-trained network; next steps are real phone recordings, full-length songs at larger scale and pitch/tempo-invariant hashes.*